

5.1 Machine Learning, Neural Networks and Deep Learning

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 01 of 17

5.1 Machine Learning, Neural Networks and Deep Learning

LEARN IT

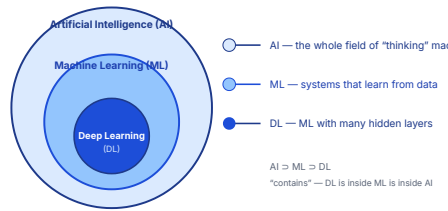
Machine learning (ML), **neural networks (NN)** and **deep learning (DL)** are important areas of **artificial intelligence (AI)**. **AI** is the broad field in which computers perform tasks that normally need human thinking, such as recognising a face, understanding speech or making a decision.

These technologies help computers **learn from data** and make useful decisions **without being explicitly programmed for every single step**. You do not write a separate rule for every possible face or every Urdu accent. You show the computer many examples, and it finds the patterns itself.

They are used in many modern systems:

- **Face unlock** on smartphones.
- **Speech recognition** (for example Siri, Alexa, or Urdu voice typing on a phone).
- **Photo search** categorisation (typing "car" in a gallery and seeing car photos).
- **Medical diagnosis** from X-rays and scans.
- **Language translation** (English to Urdu or Sindhi).

Remember the nesting: **AI is the biggest set**. Inside AI sits **ML**. Inside ML sits **DL**. Every deep-learning system is a kind of machine learning, and every machine-learning system is a kind of AI — but not every AI system is deep learning.



The set relationship $AI \supset ML \supset DL$. Deep learning is a method of machine learning; machine learning is a part of artificial intelligence.

WORKED EXAMPLE

Worked example: place each system in the correct set. Sara in Karachi uses three features on her phone.

1. **A chess app that follows fixed if-then rules** written by a programmer ("if the queen is attacked, move it"). This is **AI** (the machine decides), but it is **not ML**, because it does not learn from data.
2. **A JazzCash fraud filter** that was trained on thousands of past transactions and then flags a new unusual payment. This is **ML** (and therefore also AI). If it uses only a simple model with few layers, it may **not** be DL.
3. **Face unlock** that studied millions of face images through many hidden layers. This is **DL** — and therefore also ML and AI.

Exam sentence you can write: *All DL is ML, all ML is AI, but not all AI is ML, and not all ML is DL.*

REMEMBER IT

Russian-doll memory trick. Picture three boxes. The outer box is **AI**. Inside it is a smaller box **ML**. Inside that is the smallest box **DL**. You cannot take DL out of ML, and you cannot take ML out of AI.

COMMON MISTAKE

Mistake: "AI, ML and DL are three different, separate subjects."

Fix: They nest. $DL \subset ML \subset AI$. A deep-learning face-unlock system is all three at once.

اردو میں سمجھیں

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔ اس کے اندر مشین لرننگ (ML) ہے، اور مشین لرننگ کے اندر ڈیپ لرننگ (DL) ہے۔ یہ ٹیکنالوجیز کمپیوٹر کو ڈیٹا سے سکھاتی ہیں تاکہ ہر قدم الگ الگ پروگرام کیے بغیر فیصلے کیے جا سکیں، جیسے چہرے سے فون کھولنا یا آواز پہچاننا۔

CHECK YOURSELF

1. Which is the largest set: AI, ML or DL?
2. Is every neural-network face-unlock system an example of AI? Why?
3. Give one modern system from this section that uses these technologies.

Answers: 1. AI ($AI \supset ML \supset DL$). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.

Next lecture: 5.1.1 Machine Learning (ML)